Project Proposal: Cross-Regional Weather Dynamics in Vietnam

Data Science and Data Visualization

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1. Overview and Research Questions

Vietnam has very different weather across the country: cold winters in the North, heavy storms in the Central region, and warm weather year-round in the South. This project analyzes 10 years of daily weather data to compare weather patterns in Hanoi, Da Nang, and Ho Chi Minh City.

Research Questions:

- How do temperatures and daily temperature changes compare across the three cities?
- How do rainy seasons differ in timing, length, and the amount of heavy rainfall?
- Are hot days (above 35°C) or heavy storm days becoming more frequent over time?

2. Intended Dataset and Key Variables

Data Source

- **Source:** Open-Meteo Historical Weather API
- **Time Range:** 10 years of daily data (e.g., 2015 to 2025)
- **Locations:**

  Hanoi: 21.0285° N, 105.8542° E

  Da Nang: 16.0544° N, 108.2022° E

HCMC: 10.8231° N, 106.6297° E

Key Variable

- **Temperature:** `mean_temperature`, `max_temperature`, `min_temperature` (°C); daily average, highest, and lowest temperatures.
- **Apparent Temperature:** `mean_apparent_temperature`,`max_apparent_temperature`, `min_apparent_temperature` (°C); "feels-like" heat index.
- **Rainfall:** `precipitation_sum` (mm) and `precipitation_hours` (hours); daily rain amount and duration.
- **Wind:** `max_wind_speed` (km/h), `min_wind_speed` (km/h), and `dominant_wind_direction`; winds for storm tracking and supporting the question included in the apparent temperature declaration.
- **[Humidity:** `mean_relative_humidity` (%); daily moisture levels in the air.
- **Sun & Daylight:** `sunshine_duration` (Seconds), `daylight_duration` (Seconds); total bright sunshine time, time between sunrise and sunset.**]**

  Though the humidity and sun&daylight variables don't directly answer the 3 research questions, we will still keep them as they help answer the layer "Why?".

  For question 1, `sunshine_duration` (Seconds) and `daylight_duration` (Seconds) determine the solar radiation a city/province receives. This can tell the reason for the difference in mean temperature and daily temperature changes between different cities. On the other hand, `mean_relative_humidity` (%) explains how the ground retains heat and radiates thermal energy

  For question 2, `sunshine_duration` and `mean_relative_humidity` are related to each other. Combined with the time, we can create a chart illustrating concurrent changes of precipitation sum and humidity. This helps identify the start and the end of the rainy season

  For question 3, `mean_relative_humidity` is necessary for identifying the heat index on hot days (>35°C). Meanwhile, the `sunshine_duration` is a supportive variable to explain why a day has a storm

- **Weather**: `weather_code` (Code 0–95); A single standard number for the day's weather (e.g., 0 = clear sunny, 61 = rain, 95 = thunderstorm).

3. Data Science Approach & Workflow

1. **Clean & Prepare:** E.g. Check for missing dates, convert dates into year/month columns, and calculate daily temperature differences (Max Temp minus Min Temp).
2. **Analyze:** Find monthly averages, count hot days (over 35°C), count rainy days (over 50 mm), and look for trends over time.
3. **Build Dashboard:** Use HTML & CSS for hosting a place to display visualizations, and D3.js so users can select a city and compare years interactively.

4. Proposed Data Visualizations

4.1 Temperature Dynamics & Variations (Research Question 1)

- **Core Question:** How do temperatures and daily temperature changes (diurnal ranges) compare across Hanoi, Da Nang, and Ho Chi Minh City?
- **"Why is that so?" Questions:**
    - Why does Hanoi experience massive seasonal temperature swings while HCMC stays warm year-round?
    - Why do daily temperature swings ($MaxTemp - MinTemp$) vary significantly by season and region?

A. Primary Visualizations (Answering RQ1)

- **Multi-Line Chart (Monthly Mean Temperature):** Displays 10-year monthly mean temperature trends across Hanoi, Da Nang, and HCMC on a single axis to highlight base regional climate zones.
- **Monthly Boxplots (Daily Temperature Ranges):** Visualizes the distribution of daily temperature ranges ($MaxTemp - MinTemp$) for each month and city, capturing seasonal variability and stability.

B. Explanatory Visualizations ("Why is that so?")

- **Solar Energy Dual-Axis Chart (Sunshine & Daylight Duration vs. Temperature):** Maps sunshine_duration and daylight_duration alongside mean temperature to demonstrate how incoming solar radiation drives baseline warmth and seasonal swings (especially northern winter drops vs. southern steady daylight).
- **Humidity Heat Retention Scatter Plot:** Plots mean_relative_humidity against daily temperature ranges ($MaxTemp - MinTemp$). This illustrates how high humidity in tropical HCMC locks in night heat (low daily range), whereas lower humidity days allow thermal energy to radiate back into space quickly (high daily range).

4.2 Rainy Season Timing & Intensity (Research Question 2)

- **Core Question:** How do rainy seasons differ in timing, length, and heavy rainfall amounts across the three cities?
- **"Why is that so?" Questions:**
    - What triggers the onset and termination of rainy seasons in each region?

    - Why does Da Nang's rainy season occur much later in the year (autumn/winter) compared to Hanoi and HCMC (summer)?

A. Primary Visualizations (Answering RQ2)

- **Calendar Heatmap:** Maps daily precipitation_sum across a grid of days and years for each city, making monsoon timing, length, and heavy rain clusters instantly recognizable.
- **Grouped Monthly Precipitation Bar Chart:** Shows long-term 10-year monthly averages of total rainfall to contrast seasonal peaks (e.g., Central monsoon peak vs. Southern wet season).

B. Explanatory Visualizations ("Why is that so?")

- **Synchronized Precipitation-Humidity Trend Chart:** Overlay of precipitation_sum and mean_relative_humidity over a continuous annual axis. This pinpoints atmospheric moisture build-up prior to monsoon onset and drops post-season, explaining the moisture conditions required for rain.
- **Interactive Hourly Breakdown Subgraph (Drill-down):** Clicking any daily rainfall bar triggers a detailed view linking precipitation_sum with precipitation_hours. This explains whether rain fell as brief, intense downpours (high rain amount, low duration) or persistent drizzle (moderate rain amount, high duration).

4.3 Extreme Weather & Climate Trends (Research Question 3)

- **Core Question:** Are hot days (>35°C) or heavy storm days (>50 mm) becoming more frequent over time?
- **"Why is that so?" Questions:**
    - Why does actual human body stress (heat index) feel significantly worse than air temperature suggests?
    - What atmospheric conditions (wind, humidity, solar exposure) push normal days into severe storm or extreme heat events?

A. Primary Visualizations (Answering RQ3)

- **Annual Extreme Trend Lines:** Time-series line chart tracking the 10-year trajectory of extreme events: hot days ($Max\ Temp\ >\ 35^{\circ}C$) and storm/heavy rain days ($PrecipitationSum > 50mm$).
- **Threshold Frequency Bar Chart:** Stacked/side-by-side annual count comparison per city to identify which region is experiencing the steepest increase in extreme events.

B. Explanatory Visualizations ("Why is that so?")

- **Apparent Heat Index Matrix / Heatmap:** Combines max_temperature with mean_relative_humidity to plot max_apparent_temperature ("feels-like" temperature). This explains why moderate air temperatures with high humidity yield extreme thermal discomfort compared to dry heat days.

- **Storm Driver Vector & Solar Plot:** For identified severe storm days ($> 50mm$ rain), linked charts render max_wind_speed, dominant_wind_direction, and sudden drops in sunshine_duration. This explains the meteorological mechanics behind extreme rain occurrences (e.g., typhoon/monsoons driven by strong prevailing winds).

4.4 Advanced Interactive Analytics

Visualization: Interactive Correlation Dashboard (Scatter Plot + Linked Views + Drill-down)

Users can interactively select a city, year, or season to explore relationships among weather variables.

The dashboard includes:

- Scatter plot showing **Humidity vs Temperature**, with point color representing precipitation level.
- Brushing and filtering to focus on specific months or seasons.
- Linked charts that automatically update to display sunshine duration, rainfall, and temperature distributions for the selected observations.
- Drill-down capability from monthly summaries to daily weather records.

Purpose

Rather than only displaying weather statistics, this visualization supports exploratory data analysis by allowing users to investigate **why** weather differs between regions. It reveals hidden relationships among humidity, sunshine duration, rainfall, and temperature that are difficult to identify from aggregated charts alone.

5. Expected Key Insights

- **Regional Temperature Trajectories (RQ1 Core):**
  - Hanoi will display large seasonal swings, dropping significantly during winter months due to northern monsoons, while HCMC will show near-constant high temperatures year-round. Da Nang will present a transitional middle ground.
- **Solar & Humidity Thermal Drivers (RQ1 Explanatory):**
  - Northern winter temperature dips correlate directly with diminished daylight_duration and sunshine_duration.
  - HCMC's small daily temperature range ($MaxTemp - MinTemp$) is explained by consistently high relative humidity, which retains surface heat overnight.

- **Monsoon Timing & Structure (RQ2 Core):**
    - The calendar heatmap will reveal distinct rainy season timelines: HCMC's monsoon spans May to October, Hanoi's spans summer, while Da Nang experiences a shifted, concentrated rainy season in late autumn (September to December).
- **Atmospheric Moisture Triggers (RQ2 Explanatory):**
    - A persistent rise in mean_relative_humidity above regional baselines reliably precedes the official onset of the rainy season.
    - Drill-down analytics will reveal that southern rainy days often consist of short, heavy afternoon bursts (low duration, high volume), whereas northern wet days frequently involve prolonged drizzle (high duration, moderate volume).
- **Extreme Weather Progression (RQ3 Core):**
    - Over the 10-year timeline, annual counts of days exceeding 35°C and 50mm rain are expected to show a positive slope, confirming localized climate warming/instability trends.
- **Apparent Heat & Storm Dynamics (RQ3 Explanatory):**
    - "Feels-like" temperature (max_apparent_temperature) during hot days will significantly exceed actual air temperature due to high relative humidity, explaining severe body discomfort during heatwaves.
    - Heavy rain events in Da Nang and Hanoi will strongly correlate with elevated max_wind_speed and dominant easterly/northeastern wind vectors, identifying typhoon- and monsoon-driven origins.

6. Task allocation

Member	Main Responsibilities	Deliverables	Weeks
Trần Minh Phúc (Project Coordinator & Data Engineer)	Manage GitHub repository, collect data from Open-Meteo API, preprocess dataset, maintain project schedule	Dataset, cleaned CSV/JSON files, GitHub management	Week 1-3
Eva Wimmer (Visualization Developer - RQ1)	Implement temperature-related visualizations using D3.js, interactive filtering, legends, tooltips	Multi-line chart, Boxplot, Temperature Dashboard	Week 3-6
Dương Minh Chính (Visualization Developer - RQ2)	Implement rainfall visualizations and drill-down interaction	Calendar Heatmap, Rainfall Bar Chart, Drill-down View	Week 3-6

Evgeniia Kazakova (Visualization Developer - RQ3 & UI)	Extreme weather visualizations, dashboard layout, styling, final integration	Trend Charts, Heatmap, Dashboard UI/UX	Week 3-7

Shared Responsibilities

- Literature review
- Dashboard integration
- Testing & debugging
- Presentation slides
- Final report
- Demo rehearsal

7. Project structure

```
└── weather-dashboard/
    ├── index.html
    ├── assets/
    │   ├── fav.ico
    │   └── image.png
    ├── data/
    │   └── weather.csv
    ├── js/
    │   ├── chart.js
    │   ├── utils.js
    │   └── app.js
    └── css/
        └── style.css
```

References

Open-Meteo Documentation: https://open-meteo.com/en/docs/historical-weather-api

Vietnam's Three Climate Zones: https://www.vietnamtourism.com/en/weather